

ICT702 Data Visualisation

Exploring Data Visually

Module 6



I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



Part 4

Visualising Geospatial Data

Geospatial data include information on the geographic location of each record.

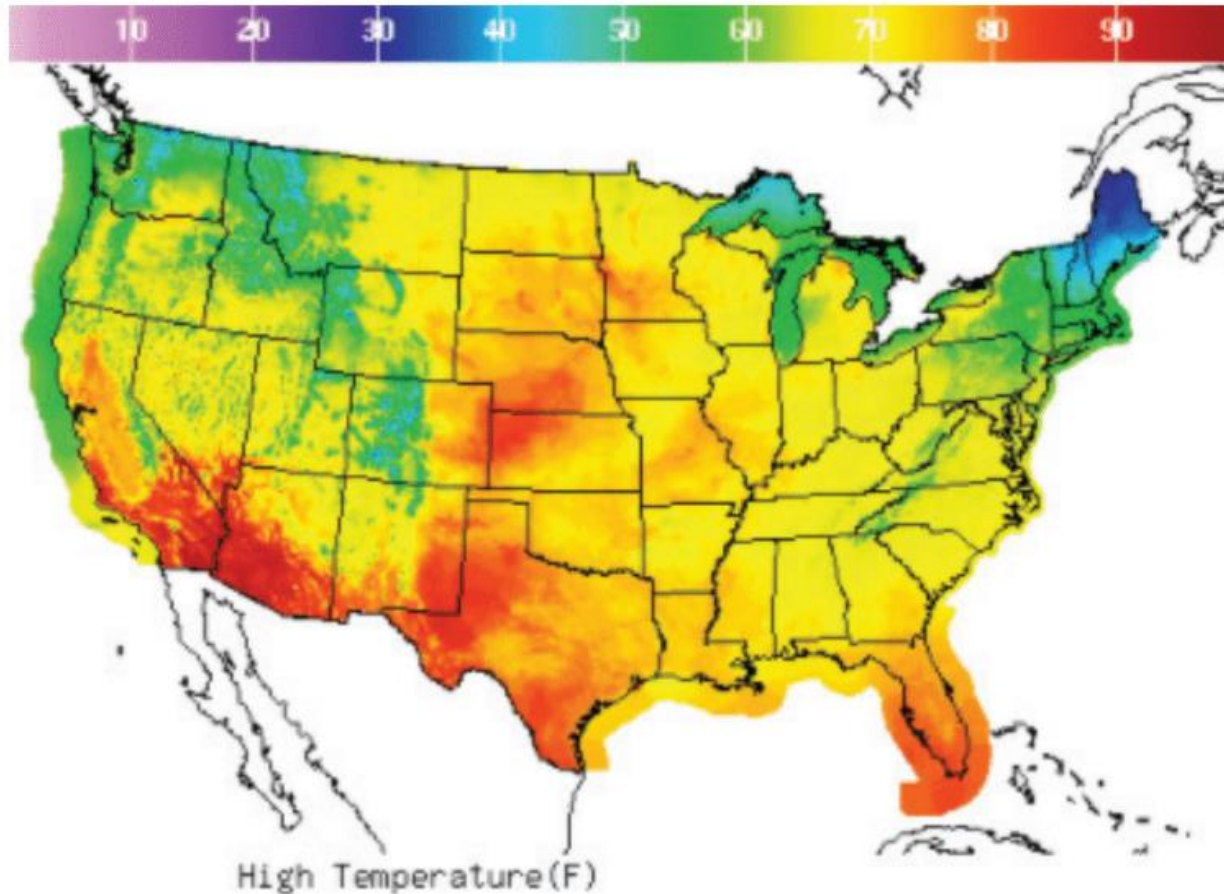
Displaying geospatial data on a map can help discover patterns that would have been otherwise difficult to interpret.

There are two general types of geographic visualisation:

- **Choropleth maps**
- **Cartograms**

A Choropleth Map Example

Weather map example

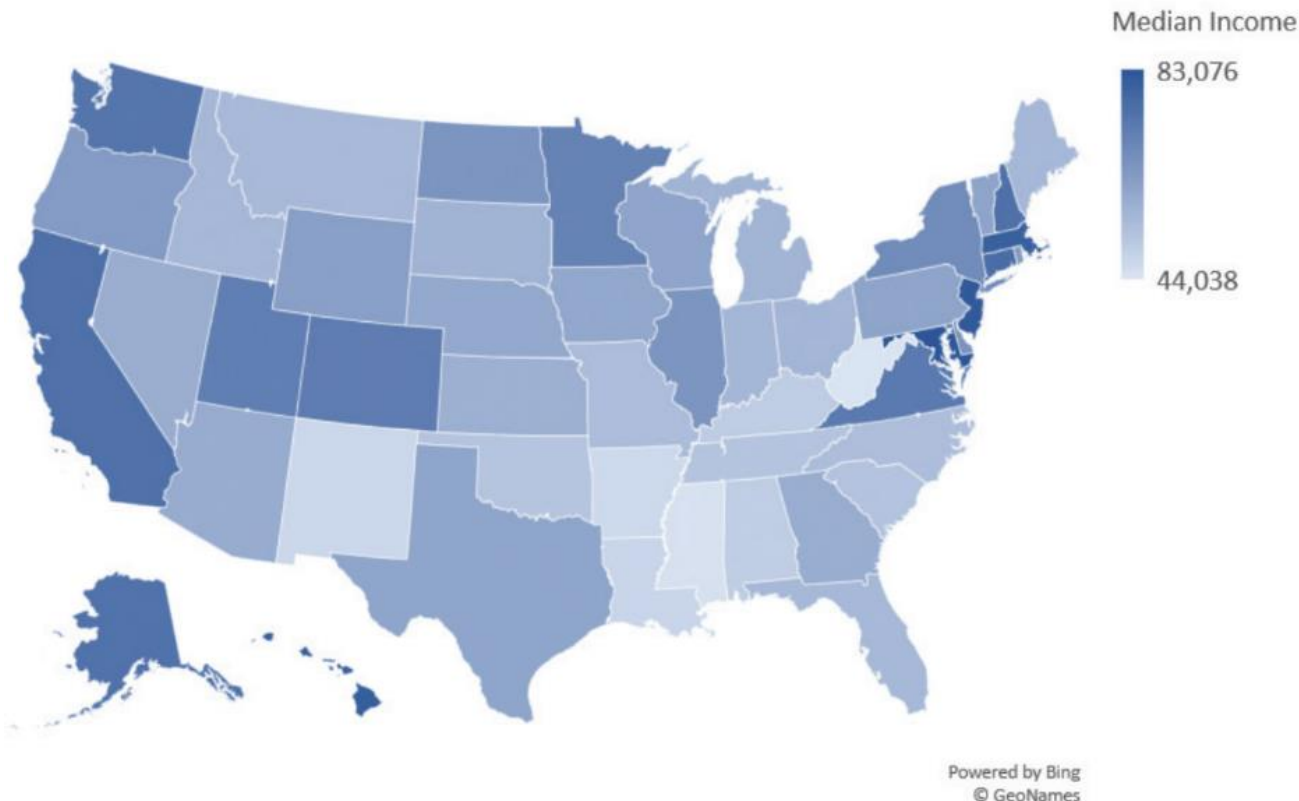


A **choropleth map** is a geographic visualisation that uses shades of colour, different colours, or symbols to indicate the values of a quantitative or categorical variable associated with a geographic region or area.

Choropleth Maps on State-Level Data

Choropleth map of the United States displaying median income by state

Median Income By State

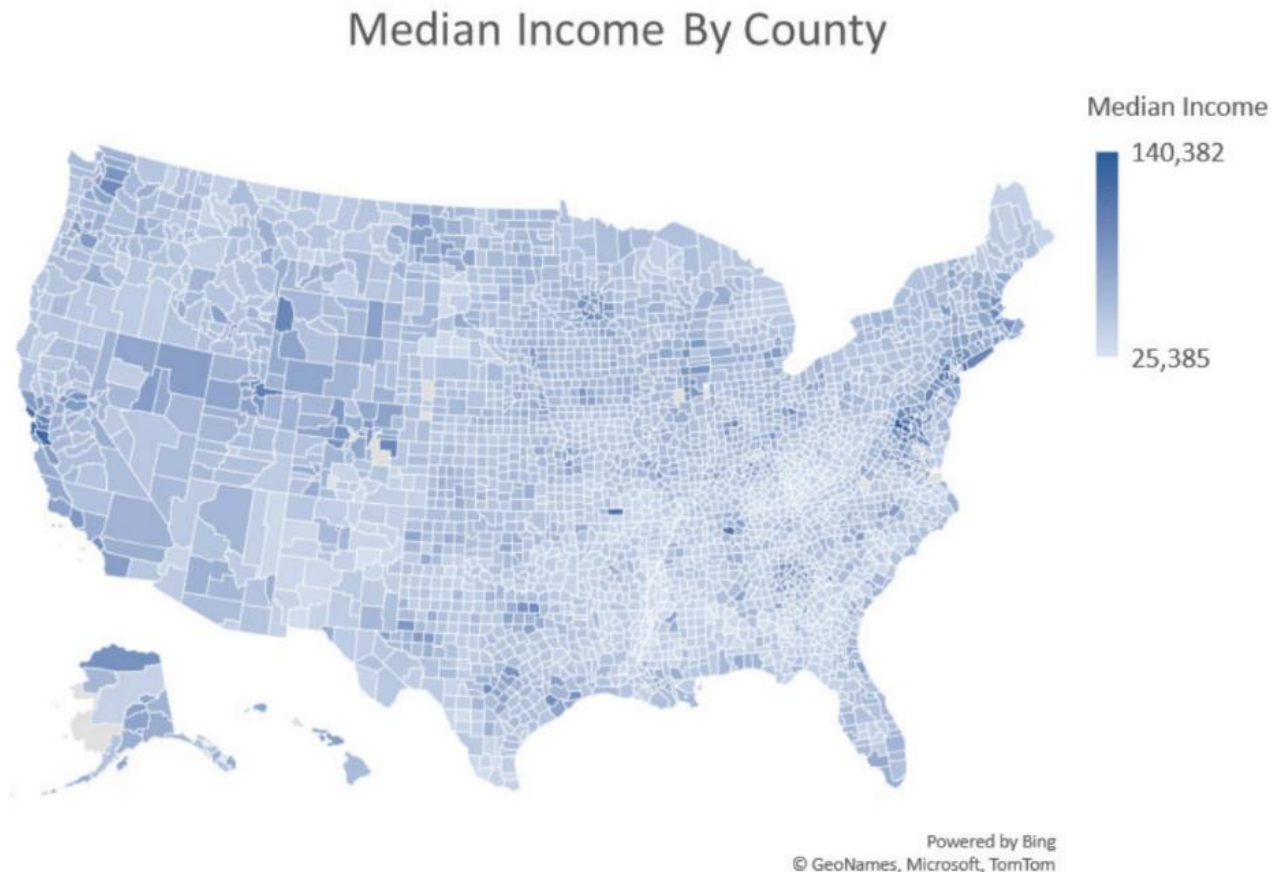


A portion of data in the file *IncomeByState*

	A	B
1	State	Median Income
2	Alabama	49,881
3	Alaska	74,912
4	Arizona	59,079
5	Arkansas	47,094
6	California	75,250

Choropleth Maps on County-Level Data

Choropleth map of the United States displaying median income by county

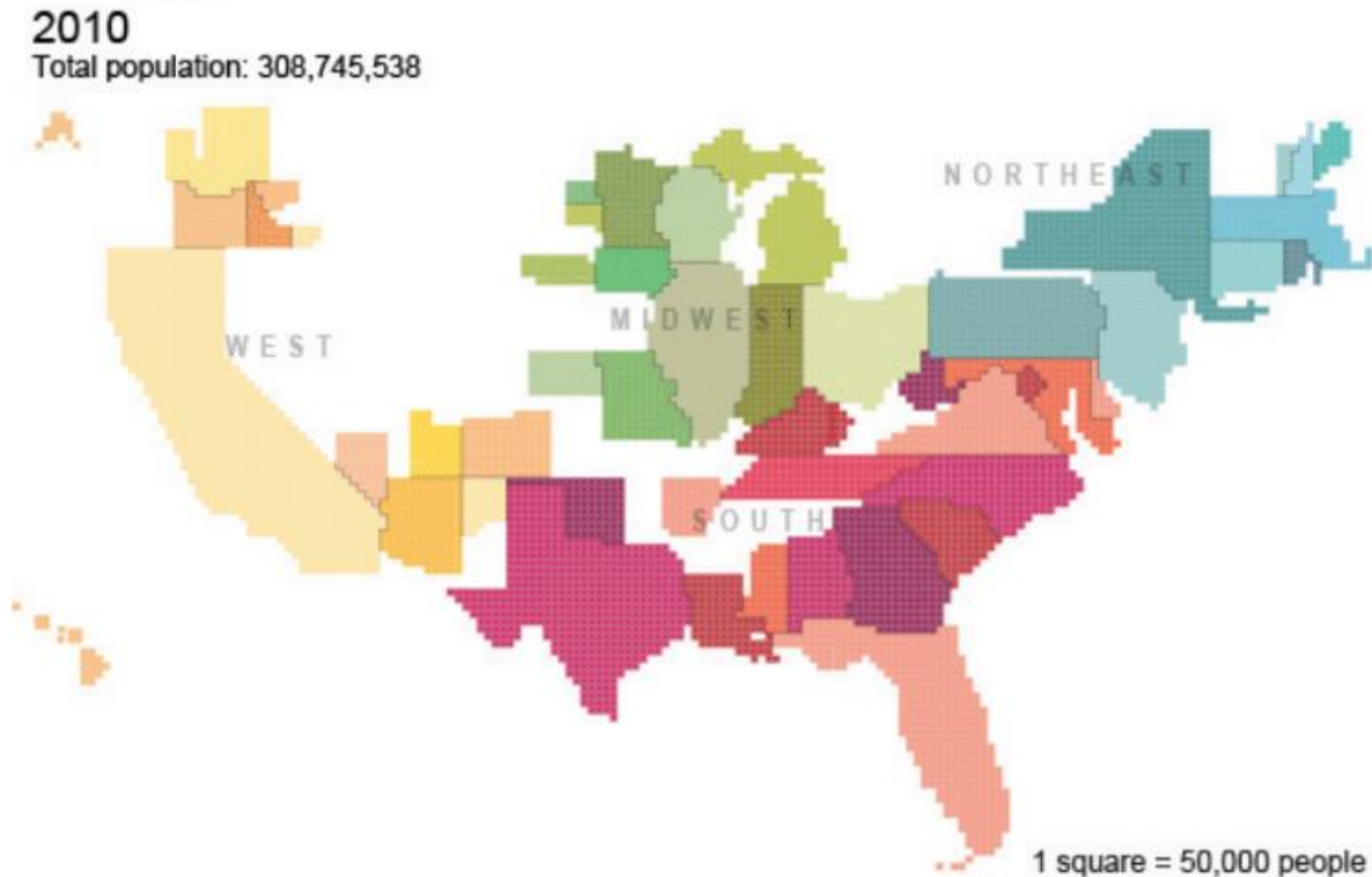


A portion of data in the file *IncomeByCounty*

	A	B	C
1	State	County	Median Income
2	AL	Autauga	59,338
3	AL	Baldwin	57,588
4	AL	Barbour	34,382
5	AL	Bibb	46,064
6	AL	Blount	50,412

Cartograms

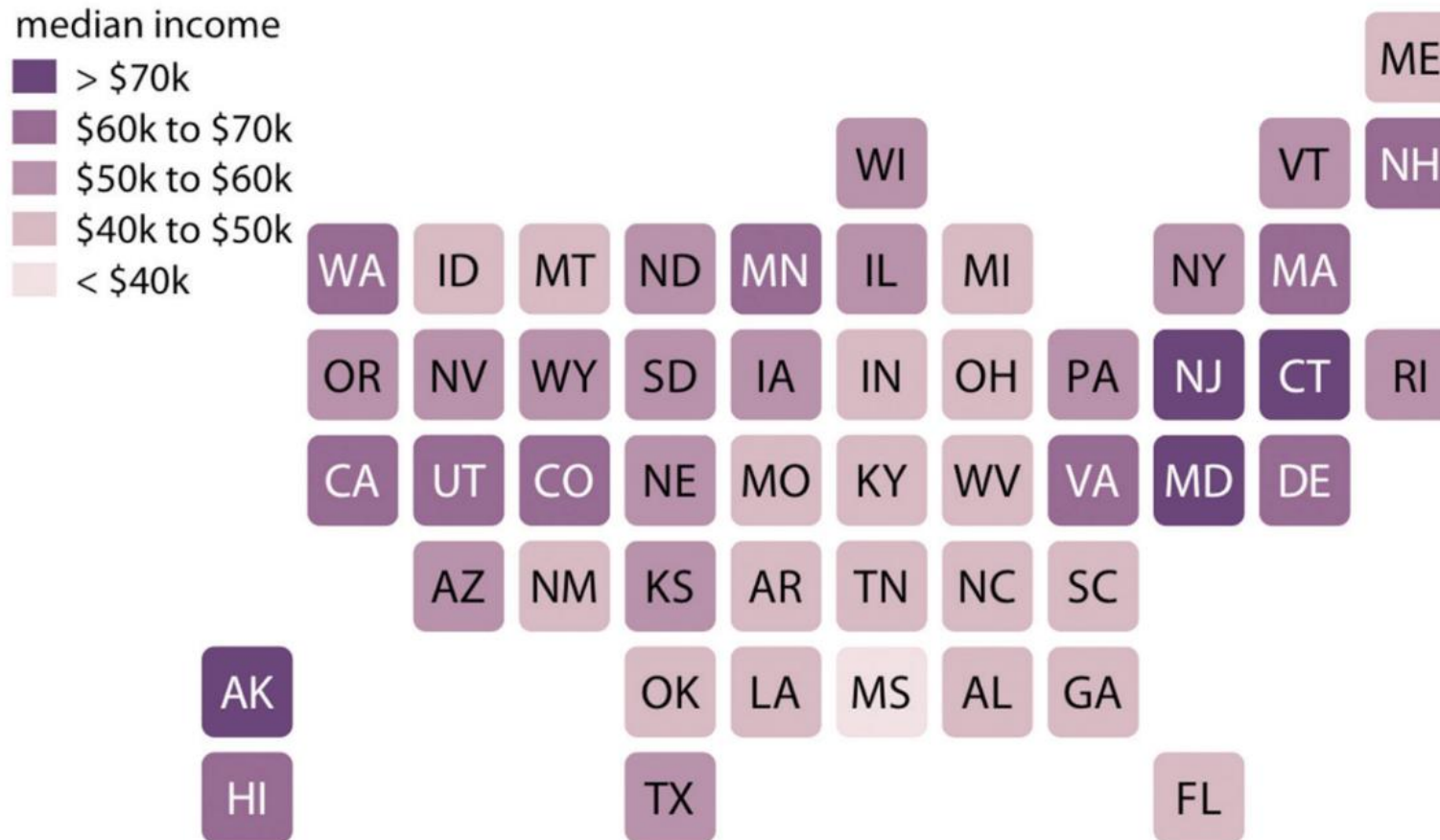
Cartogram of the United States displaying state population



A **cartogram** is a map-like diagram that uses geographic positioning but purposefully represents map regions that do not necessarily correspond to land area.

Equal-Area Cartograms

Equal-area cartogram of the United States displaying median income by state



An **equal-area cartogram** provides a balanced visual representation of each state while still maintaining fidelity to relative geographic positioning.

Summary

In Module 6 you have learnt how to:

- Discuss the visualisation techniques for conducting exploratory data analysis.
- Use Excel tools for rearranging data to facilitate the visual exploration of a data set.
- Define the types of missing data and how to potentially address missing entries.
- Explain the process of exploring data, including the distributional analysis of individual variables and the crosstabulation of multiple variables
- Visualise the association between pairs of quantitative variables with scatter charts, and with the consideration of a categorical variable.
- Visualise the pairwise association between quantitative variables with the scatter-chart matrix and the table lens.
- Visualise trend, variability, and seasonality on time series data using line charts.
- Discuss the strengths and weaknesses of choropleth maps and cartograms.

End of Part 4

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